

CUSTOMER DEMOGRAPHICS STUDY

Data Cleaning, SQL Analysis & Power BI Dashboard

Project Report

Python • Pandas • SQL Server • Power BI

1. Executive Summary

The Customer Demographics Study is an end-to-end data analytics project focused on understanding "customer characteristics, purchasing behavior, income patterns, spending behavior, and product preferences."

"The project follows a practical analytics workflow: data cleaning with Python and Pandas, structured analysis "with SQL Server, and interactive visualization with Power BI.

The dataset was inspected and cleaned before analysis. Checks were performed for missing values, duplicate "records, column naming consistency, appropriate data types, categorical consistency, and obvious invalid values."

"The cleaned dataset was then exported and used for SQL analysis.

SQL Server was used to answer business questions involving customer counts, age, gender, income, spending "score, purchase frequency, age-group behavior, and preferred product categories. The final Power BI dashboard "

"converts these results into an interactive analytical view using KPI cards, slicers, and multiple visualization types.

2. Project Objectives

- Clean and validate the customer dataset before analysis.
- Understand the main demographic characteristics of customers.
- Measure income, spending score, and purchase frequency.
- Compare customer behavior across age groups and gender.
- Analyze preferred product categories across demographic groups.
- Use SQL to answer business-oriented analytical questions.
- Build an interactive Power BI dashboard.
- Present the project in a portfolio-ready format.

3. Dataset & Data Preparation

The dataset contains customer-level demographic and behavioral information. The analysis uses fields such as "id, age, gender, income, spending_score, purchase_frequency, last_purchase_amount, membership_years, and "preferred_category.

Field	Purpose
id	Unique customer identifier
age	Customer age and age-group segmentation
gender	Gender distribution and income comparison
income	Income analysis
spending_score	Customer spending behavior
purchase_frequency	Purchase behavior analysis

last_purchase_amount	Recent purchase amount / value proxy
membership_years	Customer membership tenure
preferred_category	Product category preference

4. Data Cleaning

Python and Pandas were used to prepare the dataset. The objective was to make the data reliable and consistent "

"before moving to SQL Server.

Check	Result
Missing/null values	Checked; no null values were identified.
Duplicate records	Checked; no duplicate rows were identified.
Column names	Standardized to lowercase.
Data types	Inspected and confirmed as appropriate for analysis.
Category consistency	Categorical values reviewed for consistency.
Invalid/problematic values	Reviewed for obvious invalid entries.

After validation, the cleaned data was exported as a CSV and imported into SQL Server for analysis.

5. SQL Analysis

SQL Server was used to translate business questions into measurable queries. The analysis used COUNT, "COUNT/DISTINCT customer logic, AVG, ROUND, GROUP BY, CASE expressions, ORDER BY, TOP 1, and calculated metrics.

5.1 Main Business Questions

- What is the total number of customers?
- What is the average age of customers?
- What is the gender distribution?
- Which age group has the highest average income?
- Which preferred category has the most customers?
- What is the average income by gender?
- Which preferred category has the highest average spending score?
- How does purchase frequency vary by age group?
- Which product categories are preferred by different age groups?
- Which demographic group represents the most valuable customers?

5.2 SQL Learning Applied

A key part of the project was learning when to use TOP 1. It is appropriate when the question asks for one "winner, such as the age group with the highest average income. It is not appropriate when the purpose is to "compare every age group. CASE expressions were used to create age groups, and GROUP BY was used to compare "those groups.

6. Power BI Dashboard

The Power BI dashboard provides an interactive summary of customer demographics and shopping behavior.

"The dashboard uses KPI cards, slicers, and a mixture of chart types to avoid repetitive visualizations.

6.1 KPI Cards

KPI	DAX Logic
Total Customers	DISTINCTCOUNT(customer_segmentation_data[id])
Average Age	AVERAGE(customer_segmentation_data[age])
Average Income	AVERAGE(customer_segmentation_data[income])
Average Spending Score	AVERAGE(customer_segmentation_data[spending_score])
Average Purchase Frequency	AVERAGE(customer_segmentation_data[purchase_frequency])

6.2 Main Visuals

- Gender Distribution — customer composition by gender.
- Average Income by Age Group — comparison of income across age segments.
- Average Purchase Frequency by Age Group — comparison of purchasing behavior.
- Preferred Category by Age Group — category preference across age segments.
- Income vs Spending Score — relationship between income and spending behavior.

6.3 Slicers

- Gender
- Age Group
- Preferred Category

6.4 Age Grouping

The dashboard uses the age groups 15–25, 26–35, 36–45, 46–55, and 56+. A numeric sort column was created so that Power BI displays the groups in logical age order rather than alphabetical order.

7. Key Business Insights

The dashboard is designed to highlight differences between demographic groups rather than relying on a single "

"overall customer average. Income, spending score, and purchase frequency provide complementary views of customer behavior.

- Income can be compared across age groups to identify higher-income segments.
- Purchase frequency provides a behavioral measure that complements demographic information.
- Preferred categories can be compared across age groups to support targeted marketing and category planning.
- Gender-based income comparisons provide an additional demographic perspective.
- Income and spending score should be treated as different measures: higher income does not automatically mean higher spending.
- The income-versus-spending scatter analysis can be used to identify patterns and potential customer segments.

Exact numeric findings should be read from the final Power BI dashboard and SQL result set, ensuring that reported values remain tied directly to the final dataset.

8. Limitations

- This is descriptive analysis and does not establish causation.
- The customer-value metric is a proxy based on available purchase-related fields and is not true customer lifetime value.
- Different age-group definitions may produce different segment comparisons.
- The conclusions are limited to the available dataset and its variables.

9. Tools & Technologies

Tool	Purpose
Python	Data inspection and preparation
Pandas	Data cleaning and transformation
SQL Server	SQL analysis and business questions
Power BI	Interactive dashboard and visualization
DAX	KPI measures and calculated fields
GitHub	Version control and portfolio presentation

10. Project Deliverables

Recommended repository structure:

```
Customer-Demographics-Study/  
├── notebooks/  
│   └── customer_demographics_data_cleaning.ipynb  
├── sql/  
│   └── customer_demographics_analysis.sql  
├── dashboard/  
│   └── Customer_Demographics_Analysis.pbix  
├── images/  
│   └── dashboard.png  
└── README.md
```

If the dataset license does not allow redistribution, do not publish the dataset in the public repository; document its source in the README instead.

11. Conclusion

The Customer Demographics Study demonstrates a complete data analytics workflow from data preparation to "business visualization. Python and Pandas were used to validate and clean the data, SQL Server was used for "structured analysis, and Power BI was used to communicate the results through interactive KPIs, demographic "segmentation, category analysis, and behavioral visuals.

The project demonstrates practical entry-level Data Analyst skills including data cleaning, SQL aggregation, "CASE-based segmentation, ranking, DAX measures, dashboard design, and business-oriented interpretation. It also "creates a strong foundation for future projects with more advanced SQL techniques such as joins, CTEs,

subqueries, "

"window functions, ranking, and cohort analysis.

12. Final Power BI Dashboard

Customer Demographics & Shopping Behavior Analysis

Final interactive dashboard preview showing customer KPIs, demographic filters, category preferences, purchase frequency, and income–spending relationships.

